



Vidyavardhaka College of Engineering

Affiliated by VTU and approved by AICTE Accredited by NAAC & NBA with 'A' grade
Gokulam 3rd stage, Mysuru , Karnataka.



Department: CSE(Artificial Engineering & Machine Learning)

Audio-Visual Speech Enhancement System for Hearing Impaired and Deaf Users in Multi-Speaker Environments

TEAM MEMBERS

Akash J
Dhanush H S
Vikas B K

UNDER THE GUIDANCE OF

Prof Seema H R
Assistant Professor
Dept. of CSE(AI&ML), VVCE

Problem Statement

In environments where multiple people speak at the same time, it becomes very difficult to understand a single speaker clearly. This is widely known as the "Cocktail Party Problem."

Systems that rely only on audio signals are not sufficient in such conditions. For deaf and hearing impaired individuals, lip movements serve as a source of information for understanding speech.

However, existing Audio-Visual (AV) systems have several limitations:

- They can process only one speaker at a time
- They fail when the speaker's lips are not clearly visible
- They require high computing power, making them unsuitable for mobile devices
- They are mostly trained on Western English data and do not perform well in Indian noise environments

This project aims to address all of these limitations in one unified system.

Literature Survey

Sl no	Author and Published Year	Paper Title	Methodology Used	Problem Solved	Advantages	Inference and Limitations
1	Michelsanti et al., 2021	An Overview of Deep-Learning-Based Audio-Visual Speech Enhancement and Separation	A comprehensive survey that reviews and compares a large number of AV speech enhancement methods. Groups them by how audio and video are combined - early fusion, late fusion, or a hybrid approach	What is the current state of AV speech enhancement research and where do the major gaps lie?	Clearly maps the progress made across the field. Identifies lip-audio synchronisation and fusion strategy as the two most important factors for performance	Not a working system. Confirms that performance drops significantly in real-world settings when lips are occluded or not clearly visible
2	Adeel et al., 2019	Lip-Reading Driven Deep Learning Approach for Speech Enhancement	A lip-reading module analyses lip movements and estimates the probability of each speech sound (phoneme). These probabilities are fed into a Wiener filter — a signal processing method — to remove noise and enhance the speech signal	How to enhance speech specifically for people with hearing difficulties in real-world noisy places like cafés and streets	One of the very few papers designed specifically for hearing-impaired users. Tested in actual noisy environments. Showed measurable improvement in both speech quality (PESQ) and speech clarity (STOI) scores	Limited to one speaker at a time. Too computationally heavy for phones or wearable assistive devices
3	Rahimi et al., 2022	Reading to Listen at the Cocktail Party	A Transformer-based model that uses three types of input together — audio, lip video, and text transcription. Designed to handle situations where one or more of these inputs are missing or out of synchronisation	How to separate speech reliably even when some input modalities are unavailable or out of sync	More robust to real-world conditions than earlier systems. Works reasonably well even when one modality is missing	Handles only one target speaker at a time. Not real-time. Not designed for deaf or hearing-impaired users

Literature Survey

Sl no	Author and Published Year	Paper Title	Methodology Used	Problem Solved	Advantages	Inference and Limitations
1	Ephrat et al., 2018	Looking to Listen at the Cocktail Party	A neural network that takes mixed audio and the face video of a target speaker as input, focuses on lip movements to separate the target voice from background noise. Trained on AVSpeech - a large dataset of video clips	How to separate one person's voice from a noisy mix of multiple speakers using visual information	Works without prior knowledge of the speaker. Showed that lip movements alone are a strong cue for voice separation. Outperformed all audio-only methods at the time	Cannot handle occluded or poorly lit lips. Processes only one speaker at a time. Not designed for real-time use or for deaf users specifically
2	Afouras et al., 2022	Deep Audio-Visual Speech Recognition	Transformer-based models trained for audio-visual automatic speech recognition (AV-ASR). Uses both audio features and lip movement features to recognise speech even in very noisy conditions	How to achieve reliable speech recognition in noisy environments using both audio and visual inputs	Showed that a visual-only model can already achieve good recognition rates. Combining audio and visual inputs gives very strong performance even at very high noise levels. The lip-reading features from this work are directly useful for enhancement systems	The focus is on speech recognition, not speech enhancement. Requires significant computing power — not suitable for edge devices
3	Hou et al., 2018	Audio-Visual Speech Enhancement Using Multimodal Deep Convolutional Neural Networks (AVDCNN)	Two separate CNN branches - one processes audio, one processes video - run in parallel. Their outputs are merged before final speech reconstruction. The model also tries to reconstruct the original video as an additional task	How to improve speech clarity by combining audio and visual information in a structured way	Showed consistent improvement over audio-only models. Proved that parallel audio-visual processing is an effective approach even in simpler architectures	Handles only one speaker at a time. No mechanism to handle blocked or missing lip views. Cannot run in real-time

Literature Survey

Sl no	Author and Published Year	Paper Title	Methodology Used	Problem Solved	Advantages	Inference and Limitations
1	Ma et al., 2025	Real-Time Audio-Visual Speech Enhancement Using Pre-Trained Visual Representations	Uses visual features pre-trained from lip-reading and speaker recognition tasks. These features are integrated into a causal streaming network - meaning it processes audio and video continuously without needing to see future frames. Runs on a standard CPU	How to bring AV speech enhancement into real-time operation on everyday hardware	Processes speech with under 50 milliseconds of delay on a regular CPU. Practical enough to be integrated into hearing aids or wearable assistive devices	Only partially handles multi-speaker scenarios. Does not address visual occlusion
2	Pan et al., 2020	MuSE — Multi-Modal Target Speaker Extraction with Visual Cues	A system that identifies and extracts the target speaker's voice using only their lip video — no clean reference audio clip is needed for enrollment. The model conditions its extraction process entirely on visual input	How to extract a target speaker's voice without requiring a clean audio sample from that person beforehand	Practically very useful for deaf users who cannot provide a voice enrollment sample. Performs well across multiple standard benchmarks and surpasses earlier methods on standard evaluation metrics	Only partially handles occlusion. Not capable of real-time operation
3	Pan et al., 2024	RAVSS — Robust Audio-Visual Speech Separation in Multi-Speaker Scenarios with Missing Visual Cues	Uses an occlusion detection module that continuously monitors whether the speaker's lips are visible. When lips are visible, the system uses visual guidance. When lips are blocked, it automatically switches to audio-only processing. Evaluated on VoxCeleb2 and LRS3 datasets	How to separate multiple speakers simultaneously while still working correctly when lip visibility is lost	First system to reliably handle both multi-speaker separation (up to 5 speakers) and visual occlusion in one model. Consistent improvement over all previous baselines	Not capable of real-time processing. Not specifically optimised for deaf or hearing-impaired users

Research Gaps Identified

- **Multi-speaker limitation:** The majority of existing systems are designed to handle only one target speaker at a time. Separating multiple speakers simultaneously remains a largely unsolved problem. RAVSS is the only system that addresses this, and only up to five speakers.
- **Visual occlusion:** Most systems assume that the speaker's lips are always clearly visible and well-lit. In real-world conditions - such as when a speaker turns their head, wears a mask, or is in poor lighting - nearly all existing systems fail to perform reliably.
- **Lack of focus on assistive technology:** Despite being directly relevant to deaf and hard-of-hearing users, most of the reviewed systems are built for general noise reduction. None of them, except Adeel et al., specifically optimise for the intelligibility metrics that matter most to hearing-impaired users.
- **High computational requirements:** Most high-performing models depend on server-grade GPUs to run. This makes them completely impractical for use on smartphones, which is where hearing-impaired users actually need this technology.
- **Lack of Indian language and noise data:** All standard benchmarks use native English speakers recorded in Western environments. Indian-accented English and typical Indian urban noise environments - such as markets, public transport, etc., are not represented in any of the reviewed datasets.

Finalized Objectives

- Build a system that can cleanly separate 2 to 5 people talking at the same time in a noisy environment
- Make it smart enough to handle situations where the speaker's lips aren't visible - switching to audio-only mode automatically when needed
- Get it running in real-time on a regular CPU - no GPU required, so it can actually work inside a phone
- Design it with deaf and hard-of-hearing users at the centre - optimising for how understandable speech actually sounds to them.
- Test it on Indian-accented English in real Indian environments - cafés, metro stations, markets

Proposed Methodology

- **Speaker Identification without Audio Enrollment (from MuSE)** - The target speaker is identified using their lip video alone. No clean reference audio sample is required, which makes the system accessible to deaf users who cannot provide one.
- **Multi-Speaker Separation (from RAVSS)** - The separation backbone supports up to 5 simultaneous speakers, addressing the core multi-speaker limitation found across most existing systems.
- **Occlusion Handling (from RAVSS)** - An occlusion detection module monitors lip visibility in real time. If lips become blocked or unclear, the system automatically falls back to audio-only processing without any interruption.
- **Real-Time Streaming (from RAVEN)** - Using pre-trained visual features and a streaming-friendly architecture, the whole should run with under 50ms delay on a standard CPU.
- **Evaluation on Indian Data (inspired by Adeel et al.)** - The system will be tested with Indian-accented English speakers in real Indian noise environments to ensure it performs reliably for the intended user group.

References

1. A. Ephrat et al., "Looking to listen at the cocktail party," ACM Trans. Graph., 2018.
2. J.-C. Hou et al., "Audio-visual speech enhancement using multimodal deep CNN," IEEE Trans. Emerg. Top. Comput. Intell., 2018.
3. A. Adeel et al., "Lip-reading driven deep learning approach for speech enhancement," IEEE Trans. Emerg. Top. Comput. Intell., 2021.
4. D. Michelsanti et al., "An overview of deep-learning-based AV speech enhancement," IEEE/ACM Trans. Audio, Speech, Lang. Process., 2021.
5. T. Afouras et al., "Deep audio-visual speech recognition," IEEE Trans. Pattern Anal. Mach. Intell., 2022.
6. Z. Pan et al., "MuSE: Multi-modal target speaker extraction with visual cues," arXiv:2010.07775, 2020.
7. A. Rahimi et al., "Reading to listen at the cocktail party," Proc. CVPR, 2022.
8. T. Pan et al., "RAVSS: Robust AV speech separation with missing visual cues," arXiv:2407.19224, 2024.
9. T. A. Ma et al., "Real-time AV speech enhancement using pre-trained visual representations," arXiv:2507.21448, 2025.